



(*Capsicum annuum*; *Capsicum frutescens*)

Chili pepper

Nutrition and cooking:

Chili pepper fruits are rich in β -carotene, vitamin C, folate and protein. Fresh or dried chili peppers are cooked with other foods as flavoring and spice. They can also be used to make chili sauce.

Characteristics:

Optimum day temperatures for chili pepper growth range from 20 to 30°C. When the temperature falls below 15°C or exceeds 32°C for extended periods, growth and yield are usually reduced. Peppers can be grown under different day lengths. Chili pepper grows best in a loam or silt loam soil with good water-holding capacity, but can grow in many soil types, as long as the soil is well-drained.

Cultivation instructions:

Fill seedling trays with sowing medium, such as peat moss, commercial potting soil, or a potting mix prepared from soil, compost, rice hulls, vermiculite, peat moss and sand. Sow one seed per cell and cover 1 cm deep. Cover the seedlings with an insect-proof net (50–60 mesh), or sow them inside a greenhouse or screenhouse. This provides shade and protects seedlings from heavy rain and pests, such as aphids and whiteflies, which transmit viruses. Seedling trays should be placed on benches (avoid touching the ground to prevent infection by soil-borne diseases and vulnerability to flooding). Upon emergence, water the seedlings thoroughly every morning.

Crop rotations should avoid sequences in which peppers are planted immediately following another solanaceous crop such as tomato, eggplant, or potato. A preceding paddy rice crop is often helpful in that the flooded soil is depleted of many soil-borne pathogens and weed seeds. The ideal seedling has 4–5 true leaves, is disease-free, stocky, and has no flowers. Use 1.5-m wide (furrow to furrow), 30 cm high raised beds which are prepared with a mixture of compost and NPK fertilizers. Rows should be 55 cm apart, with 45 cm between plants within rows. Transplant in the late afternoon or on a cloudy day to minimize transplant shock. Furrow or drip irrigation are recommended; overhead irrigation should be avoided as wet leaves and fruits



promote disease development. Pepper plants cannot tolerate flooding and fields should be drained quickly after heavy rain. Each plant is individually staked before flowering stage. Apply NPK fertilizer during the growth (3 and 6 weeks after transplanting) and harvesting periods (every 2–3 weeks).

Information source:

Berke, T., Black, L. L., Talekar, N. S., Wang, J. F., Gniffke, P., Green, S. K., Wang, T. C., and Morris, R. 2005. International cooperators' guide: Suggested cultural practices for chili pepper. AVRDC publication number 05-620.



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